

CL-1002
Programming Fundamentals
Lab # 12

Objectives:

- Practice and understanding on basic C++ programs
- 1D Arrays

Note: Carefully read the following instructions (*Each instruction contains a weightage*)

1. First think about statement problems and then write your program.
2. Write Program in C compiler/IDE and save source file **for each program**.
3. **Do not copy from any source otherwise you will be penalized with negative marks.**
4. Complete your lab **within given Time Slot**.
5. Add your source code in this word document + Make one ZIP file of your all source codes.
6. Please submit your **Both files** with this naming convention ROLLNO_SECTION_LABNO.
7. Submit your lab on Google Classroom.

Problem: 1

Write a C++ program named "**eliminate_duplicates**" that takes an array of integers and eliminates all the duplicate integers in the array.

If any duplicate integers are eliminated, then the program should also tell the new size of the array (number of distinct integers in the array). Here is an example. Suppose the following array values are input from user;

0	1	2	3	4	5	6	7	8	9	10
58	26	91	26	70	70	91	58	58	58	66

Then the program should alter the array so that it looks like this:

0	1	2	3	4	5	6	7	8	9	10
58	26	91	70	66	??	??	??	??	??	??

and it should display the new size of the array is 5 instead of 11 . The question marks in the cells after the 5th cell indicate that it does not matter what numbers are in those cells.

Problem: 2

Write a C++ program that checks if a given array is a palindrome array or not. Palindrome number or array is one which becomes the same when read from left side and right side i.e. 12321, abbcdbba etc

Remember that you are not allowed to make a second array in the process.

Sample Input in array-1: ['a', 'b', 'c', 'd', 'e', 'e', 'd', 'c', 'b', 'a']

Sample Output: Yes, array is **Palindrome**

Problem: 3

Write a C++ program that creates an array of length **30** characters, then you need to get input from user to get a series of characters in array. Program should tell how many unique values are present in the array. Also, the program should tell the **occurrence** of each **unique** value in the character array.

Problem 4

Write a computer program that creates an integer array of size 10. Asks the user to populate the array with numbers as taken input by the user (user must enter all numbers between 0-9) . Form an integer variable which has the list content numbers as elaborated by following:

Sample values in Array: [2, 3, 7, 5, 0, 0, 9, , ,]

Sample Integer: 2375009

Problem: 6

Write a C++ program that creates two arrays of **20** integers, takes input from user, then program should create 2 arrays(named 'even' and 'odd') each of size 20 integers. '**Even**' array should have all even integers from 1st array and '**odd**' array should have all odd integers from 1st array.

Sample Input: [1, 2, 3, 4, 5, 6, 6, 7, 8, 9, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13]

Sample Output: **even** = [2, 4, 6, 6, 8, 8, 4, 6, 8, 10, 12]

odd = [1, 3, 5, 7, 9, 5, 7, 9, 11, 13]



Problem: 7

Write a C++ program that creates **two** arrays of **10** integers, takes input from user, then program should create a **3rd** array of size 10 integers. **3rd** array should have top **10 biggest** integers from both arrays.

Sample Input: [1, 2, 3, 4, 5, 6, 6, 7, 8, 9], [4, 5, 6, 7, 8, 9, 10, 11, 12, 13]

Sample Output: [13, 12, 11, 10, 9, 9, 8, 8, 7, 7]

Problem: 8

Write a C++ program to that will take one input string from user without space. Then your program should remove the Vowel character from the string and print it out.

Sample Input:

HellowWorld

Output:

HllwWrld

Best Wishes